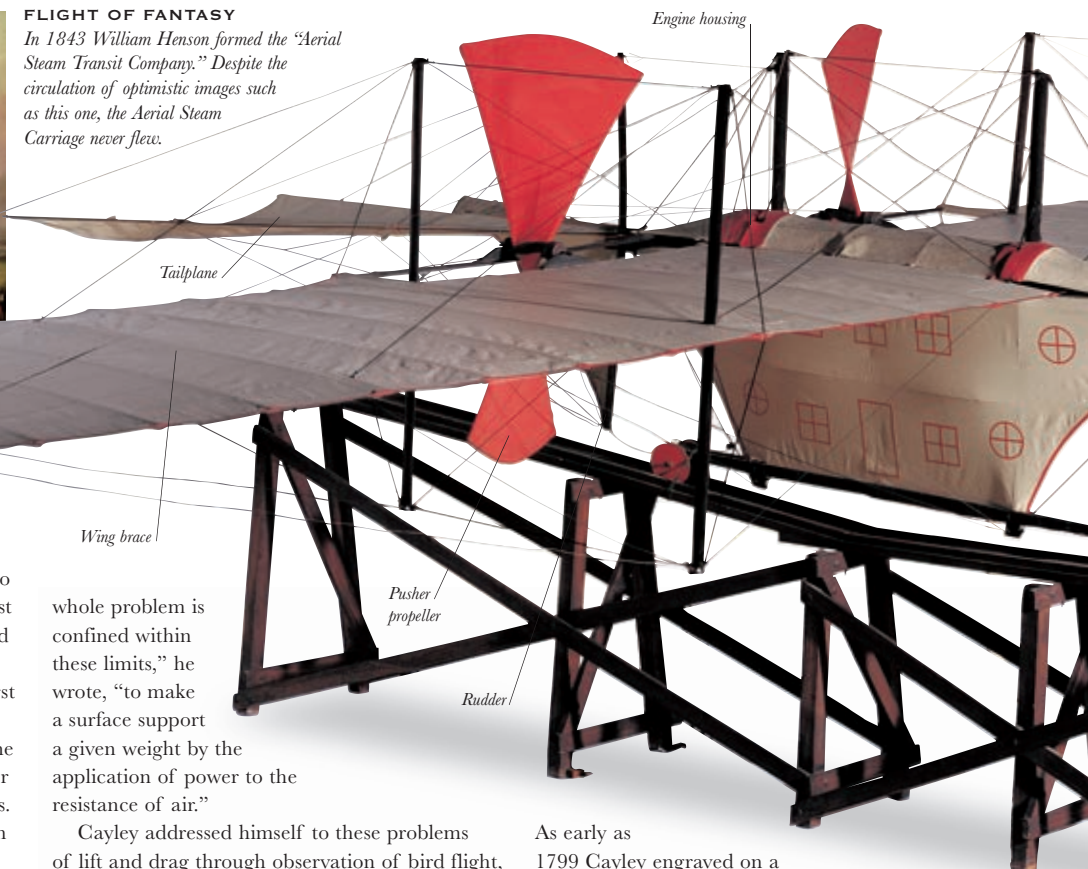




FLIGHT OF FANTASY

In 1843 William Henson formed the "Aerial Steam Transit Company." Despite the circulation of optimistic images such as this one, the Aerial Steam Carriage never flew.



But it was the Montgolfiers who established precedence and their place in the history books. As would happen in the exploration of space two centuries later, they sent animals up first on a test flight – a duck, a sheep, and a rooster. All landed safely, although according to one account "they were, to say the least, much astonished." The first free manned flight followed on November 21, when physician François Pilâtre de Rozier and the Marquis d'Arlandes, an army officer, drifted over Paris covering 5 miles (8km) in about 25 minutes. Ten days later Jacques Charles and a companion flew 25 miles (40km) in a hydrogen balloon.

Ballooning captured the public imagination much as flying machines would in the early 20th century. Crowds flocked to demonstration flights and the fliers became national heroes. French aeronaut Jean-Pierre Blanchard, with American expatriate John Jeffries on board, flew across the Channel from England to France in January 1785, 124 years before Louis Blériot (see page 41).

Scientific progress

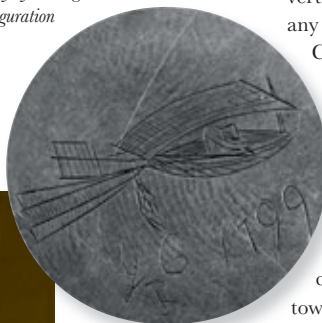
One of the many individuals fascinated and inspired by reports of the early balloon flights was a young boy growing up on his father's estate in Yorkshire, England. He was George Cayley, who was to make the first serious practical and theoretical progress toward heavier-than-air flight. Cayley could easily be considered as an eccentric – a member of the landed gentry using his privileged leisure time to pursue a fanciful hobby. But he, in fact, worked within a maturing scientific tradition, which enabled him to define precisely the challenge of heavier-than-air flight: "The

whole problem is confined within these limits," he wrote, "to make a surface support a given weight by the application of power to the resistance of air."

Cayley addressed himself to these problems of lift and drag through observation of bird flight, systematic experimentation, and mathematical calculations. He used an ingenious device known as a "whirling arm" – a precursor of the wind tunnel – to test the lift created by different airfoils, or wings, at various angles and speeds.

IMAGE OF THE FUTURE

Sir George Cayley engraved this image of a flying machine on a silver disk in 1799. Cayley's design was the first to resemble the configuration of a modern airplane.



SIR GEORGE CAYLEY

Basing his theories on experimentation and observation, Cayley (1773–1857) pioneered the conquest of flight with his works on aeronautics.



As early as 1799 Cayley engraved on a silver disk an image of a flying machine that marked a crucial step forward in design from Leonardo-style ornithopters: the wing had ceased to be the means of propulsion, becoming instead purely a device to generate lift. Through the next decade he built both model and full-size gliders. His full-size glider had a wing attached to the front end of a pole, and at the rear of the pole a vertical rudder and horizontal tailplane. "When any person ran forward in it with his full speed,"

Cayley wrote, "taking advantage of a gentle breeze in front, it... would frequently lift him up and convey him several yards together."

In 1809–10 Cayley made the results of his work public in a three-part paper, "On Aerial Navigation." His calculations of lift and drag, and his comments on how an aircraft could be stabilized and controlled, constituted a solid basis for potential progress towards heavier-than-air flight. Unfortunately, they were largely ignored. As Cayley himself admitted, flight remained "a subject rather ludicrous in the public's estimation."

The awakening of a more sustained interest in heavier-than-air flight did not come for another 30 years. It was provoked by the success of the steam engine applied to transportation systems. By the 1840s, railroad construction was booming